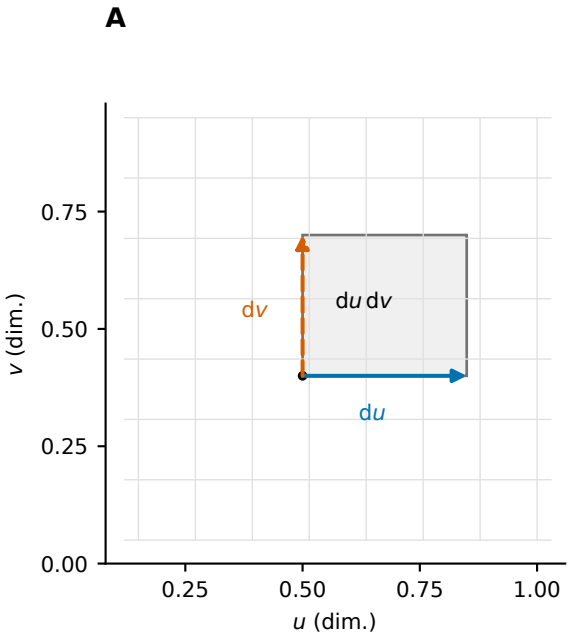
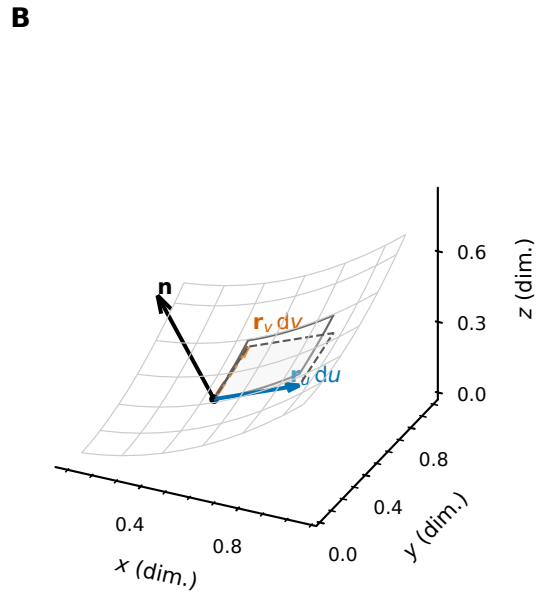


Parameter rectangle at $(u, v) = (0.5, 0.4)$



$dS = \|\mathbf{r}_u \times \mathbf{r}_v\| du\, dv$; $d\mathbf{S} = \mathbf{n}\, dS$